Rohan Batra

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Education

University of Virginia Charlottesville, VA

B.S. in Computer Science & B.A. in Mathematics, Major GPA: 3.9/4.0 *Expected May 2028*

- Relevant Coursework: Data Structures and Algorithms, Computer Systems & Architecture, Software Development, Linear Algebra, Probability, Discrete Math, Differential Equations, & Calculus I-III
- Activities: Sports Analytics @ UVA, College Mentors for Kids, Wahoops, ML @ UVA, HooHacks, Google Developer Club, Claude Developer Club

Experience

Arcturus Technologies Jun. 2024 – Aug. 2024

Software Engineering Intern *Leesburg, VA*

- Developed a **binary classification prediction** model by leveraging the **Yahoo Finance API** & **SciKit-Learn**. Trained on **OHLCV** features using **logistic regression**, improving stock price prediction accuracy by **15%**.
- Built a **backtesting engine** using **Python, NumPy, Pandas, & Matplotlib**; Integrated **Monte Carlo simulations** and **historical bootstrapping** to rigorously evaluate the performance of multiple trading strategy theses.
- Implemented comprehensive risk and statistical metrics, generating **Sharpe Ratios**, **drawdown profiles**, and **confidence intervals**, reducing strategy performance analysis time from **hours to mere minutes**.

Mathnasium Aug. 2022 – Jun. 2025

Tutor *Leesburg, VA*

- Elevated academic performance for **200+** students in subjects up to **Calculus II** by identifying core knowledge gaps and breaking down complex problems into something intuitive and understandable for them.
- Maximized student success rates and retention by continuously assessing academic progress. Increased **over 100 student's exam grades by 20+%** by deploying individualized teaching methods.

Projects

Course Spot Bot | GitHub | *Python, Google Cloud Platform, Discord Webhooks, SIS API*

- Eliminated missed course registrations by creating a serverless Python microservice that tracks specific classes/sections via the **UVA SIS REST API**.
- Delivered instant class availability alerts with **under 10-second** latency by parsing JSON API payloads and engineering a custom notification system (Discord, email, etc).
- Achieved **24/7** autonomous system uptime and eliminated manual page refreshes by deploying the application via Google Cloud Run and Cloud Scheduler on a strict **3-minute** cron interval.

JSON Validator | GitHub | *C++*

- Lightweight, robust **JSON validator** in **C++** capable of parsing and verifying the structural integrity of JSON data strings against standard formatting rules.
- Implemented a **lexer and recursive parser** to accurately tokenize and validate JSON objects, arrays, and primitive data types, ensuring optimal execution performance.

Music Agenda | GitHub | *Swift, SwiftUI, Apple Music API*

- Developed a native **macOS application** to catalog album progression & completion, successfully handling **concurrent artist and album queries** achieving minimal search latency via the **Apple Music API**.
- Enhanced music library management for users, increasing tracking accuracy by **50%**, by implementing a three-stage workflow featuring interactive progress bars, personalized notes, and a widget-app access system.

Activities

Jane Street Mystery Planet – Participant Jul. 2026

- Selected for an invite-only puzzle competition and market simulation at Jane Street's NYC office; **placed 4th**.
- Collaborated in a team to solve logic and pattern-recognition challenges; Developed trading strategies by reasoning about price discovery, market dynamics, and participant incentives.

Estimathon – Participant Dec. 2025

- Placed **5th out of 20** teams in **Jane Street**–sponsored Estimathon by collaborating with my team to solve **13 Fermi estimation** challenges in **30 minutes**.

National Computer Science Honor Society – Social Media Manager & President Aug. 2023 – Jun. 2025

- Led **3 teams to top-15 finishes (out of 30+)** at UVA's annual coding competition.
- Increased club engagement by **30%** by designing and hosting hackathons and programming challenges throughout the year.
- Developed practical coding skills for **40+** members by fostering a collaborative environment blending competition with peer-to-peer learning.

Technical Skills

Languages: Python, Java, C/C++, HTML/CSS, SQL, MATLAB, LaTeX
Frameworks & Libraries: React, Node.js/Express, Pandas, NumPy, Matplotlib
Developer Tools: Git/GitHub, Google Cloud/Apps Scripts, Firebase, JUnit Testing, REST/WebSocket, CI/CD, Agentic AI/LLMs